

Agent Trust Border

A loss-aware admission layer for agents crossing trust systems

A signed agent is still not an authorised agent

What crosses the border

Agent cards and attestations can name an identity, capability and evidence. Across ecosystems, they do not automatically preserve issuer trust, delegation, audience, revocation or receiver policy.

What the receiver needs

A deterministic border maps fields without upgrading trust, records every semantic loss, challenges missing proof, intersects exact scope and blocks every unresolved action.

One agent, three outcomes - the proof changes the verdict

01 UNKNOWN

02 ADMIT

03 DENY

Capability only

Signature valid. Deploy capability declared. Trusted grant absent. Border returns an exact nonce-bound challenge.

Exact grant

Pinned issuer binds the same request, nonce and policy. Effective scope: sandbox:demo, max_runs=1.

Replay or mutation

Reuse the nonce or request production. The binding fails; the action remains unexecuted.

The kernel never collapses trust into one VERIFIED badge

Five independent checks stay visible

Integrity, identity, authority, evidence and world truth are evaluated separately.

Passing one axis never upgrades another. Missing mandatory meaning becomes UNKNOWN, never permission.

INTEGRITY $\neq$ AUTHORITY

- Exact canonical bytes signed
- Receiver binds key to subject
- Grant intersects request + policy

EVIDENCE $\neq$ TRUTH

- Claim evidence remains attributable
- Semantic loss stays explicit
- World truth is not claimed

A falsifiable security slice, not a gateway promise

Reuse

Strict models, RFC 8785, DSSE/Ed25519, in-toto Statement, and Pramana/A2A/ACPs fixture shapes.

Proof executed

20/20 named cases passed; 20×100 fixed-input executions were byte-stable; receipt verification is offline.

Built

Offline directional crosswalk, semantic-loss ledger, exact challenge, receiver reducer, and independent receipt verifier.

Truth boundary

Synthetic pinned adapter fixtures, no real action, no model or network in the trust path, and no conformance or world-truth claim.